

Chapter 3. **GROUPS**

3.1 What is a group?

3.1.3 A taste of isomorphic groups



Today we will examine the Cayley table of some finite groups, and **conclude that such groups are ESSENTIALLY the SAME.**

A taste of isomorphic groups

A finite group is completely determined by its Cayley table since all group axioms (associativity, existence of the unity and inverses) depend entirely on the operation of the group.

Multiplicative group
 $\mathbb{Z}^* = \{1, -1\}$

$\mathbb{Z}^*$	1	-1
1	1	-1
-1	-1	1

Additive group
 $\mathbb{Z}_2 = \{\bar{0}, \bar{1}\}$

$\mathbb{Z}_2$	$\bar{0}$	$\bar{1}$
$\bar{0}$	$\bar{0}$	$\bar{1}$
$\bar{1}$	$\bar{1}$	$\bar{0}$

Notice that **these two tables are the same** in the sense that the Cayley table of $\mathbb{Z}^*$ becomes that of $\mathbb{Z}_2$ if we replace the symbols 1 and -1 by $\bar{0}$ and $\bar{1}$, respectively. Put it this way **$\mathbb{Z}^*$ and $\mathbb{Z}_2$ are the same groups except for notation**; in such a case we will say that they are **isomorphic**, or that **they are the same up to isomorphism**.

We will call

two (finite) groups isomorphic if they have the same Cayley table except for notation.

Up to isomorphism, there is only one group of order 1, 2, or 3.

- If $|G| = 1$, then $G = \{1\}$.
- If $|G| = 2$, then $G = \{1, g\}$, where $g \neq 1$ and $g^2 = 1$.
- If $|G| = 3$, then $G = \{1, g, g^2\}$, where $g \neq 1$, $g^2 \neq 1$, $g \neq g^2$ and $g^3 = 1$.

We will show that there is only one way to fill in the Cayley table to obtain a group.

For $|G| = 1$, the group reduces to the identity element $G = \{1\}$.

Next, for $|G| = 2$ we have that $G = \{1, g\}$, where $g \neq 1$. Let us fill in the Cayley table of G :

G	1	g
1	1	g
g	g	?

G	1	g
1	1	g
g	g	1

In the column corresponding to g , we have already g , so the second element of the column has to be 1.
Therefore $g^2 = 1$

To finish, let us analyse what happens when $|G| = 3$. Write $G = \{1, g, h\}$, where $g \neq h$ and both g and h are different from the identity element.

By the Cancellation laws we have that

$$gh \neq g \text{ and } gh \neq h,$$

$$(gh = g \Rightarrow h = 1!)$$

$$\text{so } \boxed{gh = 1}. \text{ Similarly } \boxed{hg = 1}$$

G	1	g	h
1	1	g	h
g	g	?	1
h	h	1	?

$$\underline{g^2 = ?} \quad \underline{h^2 = ?}$$

In the column (and row) corresponding to h we have already the elements h and 1 , so the remaining entry has to be g . This tells us that $h^2 = g$. Similarly, $g^2 = h$.

G	1	g	h
1	1	g	h
g	g	h	1
h	h	1	g

We can rewrite G as follows: $G = \{1, g, g^2\}$

and $\underline{g^3 = g g^2 = g h = 1}$

$g^2 \cdot g^2 = h^2 = g$

	1	g	g^2
1	1	g	g^2
g	g	g^2	1
g^2	g^2	1	g

IMPORTANT:

Groups consisting whose elements are all powers of one element g are called **cyclic**. We write C_n to denote the cyclic group $\{1, g, \dots, g^{n-1}\}$ of order n . We have proved that **every group of order 1, 2, or 3 is cyclic**.

CAUTION: this is NOT the case for GROUPS of ORDER 4.

$$\mathbb{Z}_2 \times \mathbb{Z}_2 = \{(\bar{0}, \bar{0}), (\bar{0}, \bar{1}), (\bar{1}, \bar{0}), (\bar{1}, \bar{1})\}$$

All the elements in $\mathbb{Z}_2 \times \mathbb{Z}_2$ have order 2.

For the tutorial:

We will see in the tutorial sessions that, up to isomorphism, **there are only two groups of order 4**, the **cyclic group C_4** and a **noncyclic group K_4** with Cayley table:

K_4	1	a	b	c
1	1	a	b	c
a	a	1	c	b
b	b	c	1	a
c	c	b	a	1

The group $K_4 = \{1, a, b, c\}$ is called the **Klein group**. The multiplication can be described as $a^2 = b^2 = c^2 = 1$ (all its elements have order 2) and the product of any two of a , b , and c is the third. Notice that $\mathbb{Z}_2 \times \mathbb{Z}_2$ and K_4 are isomorphic, while $\mathbb{Z}_4$ is isomorphic to C_4 .